

AYMANE AIT DADS

Fullstack ML Platform Engineer | Python · React · AI Infrastructure · Machine Learning Systems

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PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams in the NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, \$106K+ prize pool) by architecting an end-to-end LoRA SFT pipeline on a 30B-parameter model with BF16 precision, 8,192-token context, and custom verification solvers. Reduced manual network analysis time by 60% at Orange Maroc by building automated machine learning pipelines processing 100K+ records, with model outputs delivered directly to the network optimization team via Power BI dashboards. Brings strong Python and Linux expertise alongside experience with backend infrastructure, relational databases, REST APIs, and scalable data pipelines aligned with the ML Platform team's need to schedule, manage, and monitor machine learning experiments. Aims to contribute fullstack tooling - from Python training infrastructure to React-based dashboards - that accelerates ML engineer velocity and provides real-time visibility into experiment lifecycle at Tesla's GPU-scale autonomy platform.

CORE COMPETENCIES

Machine Learning Platform | Python Backend Infrastructure | Experiment Lifecycle Tracking | React Dashboard Development | Scalable AI Workflows | Relational Database Experience | ML Experiment Monitoring | GPU Cluster Infrastructure

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built automated machine learning pipelines (Random Forest + K-Means) on **100K+ fiber-optic network records**, reducing manual analysis time by ~60% through feature engineering and model evaluation scripts.
- Designed a clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, with outputs consumed directly by the network optimization team for infrastructure maintenance decisions.
- Developed **Power BI dashboards** translating model outputs into actionable maintenance recommendations, providing real-time performance insights to stakeholders and engineering leadership.
- Delivered structured stakeholder presentations aligning **data science model outputs** with business decisions across network quality classification for thousands of fiber-optic nodes.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge

Kaggle Competition · In Progress (June 2026) · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) by architecting a full LoRA SFT pipeline on a **30B-parameter Mamba-Transformer** (BF16, completion-only loss, 8,192-token context, r=32 targeting all attention + MLP + lm_head projections) using Unsloth and SFTTrainer.
- Ran **controlled ablation experiments** across LR schedules, synthetic augmentation strategies, and packing configurations - discovering reasoning-style consistency outweighs data volume in LLM SFT - and built custom cipher parsers and 8-bit operation solvers to produce verified chain-of-thought training data.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Triton · Kaggle GPU

AI-Generated Image Detection

EURECOM ImSecu + NTIRE 2026 @ CVPR · 1st in Class Leaderboard · 0.791 AUC

- Ranked **1st in the EURECOM class private Kaggle leaderboard** (0.791 AUC) by fine-tuning CLIP ViT-L/14 (428M parameters) with a custom MLP head, dual learning rates, and FP16 mixed precision training on 250K images across 25+ generator types.
- Submitted to the **NTIRE 2026 @ CVPR CodaBench international benchmark**; improved out-of-distribution generalization by discovering that 1-epoch training (0.793 AUC) outperformed 4-epoch training (0.767 AUC), validated through 10-view TTA ensemble weighted by validation AUC.

PyTorch · OpenCLIP · ViT · AdamW · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-Level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Cloud Computing, Web Applications, Computer Vision, NLP, Statistics, Image Security

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

ML Frameworks & Engineering: Python | PyTorch | Scikit-learn | Hugging Face Transformers | Unsloth | TRL | PEFT | vLLM

Backend, Data & Infra: SQL | PostgreSQL | REST APIs | Supabase | Docker | Linux | Git | Bash

LLM Fine-Tuning & MLOps: LoRA | QLoRA | SFTTrainer | BF16/FP16 Mixed Precision | Ablation Studies | Power BI | Kaggle GPU